

Unlikely Intersections in Crypto Exchange Reserves: An O-Minimal Test for Systemic Risk

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Abstract

Five bankruptcies within seven months across the centralized-crypto perimeter exposed a gap in systemic risk measurement: most venues lack liquid equity, so equity-based indices cannot detect joint distress. This paper builds a supervisory diagnostic from public reserve compositions. Embedded in a product of simplices, simultaneous distress becomes an unlikely intersection in the o-minimality programme. A Pila–Wilkie polylog upper bound and a heuristic power-of-horizon lower rate yield a horizon-dependent crossover, an empirically testable prediction. Under a domain-prior threshold, the full three-venue intersection is realised exactly at the spot-Bitcoin ETF approval quarter; a stablecoin placebo does not move. The diagnostic delivers a real-time audit line on public data and an illustrative state-contingent capital buffer.

JEL Classification: C14, C58, G21, G28.

Keywords: O-minimal structures; Pila–Wilkie counting; unlikely intersections; systemic risk; cryptocurrency exchanges; proof of reserves; contagion; natural experiments.

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1 Introduction

Cryptocurrency exchanges are among the most transparent counterparties in modern finance: several of the top-ten venues by customer assets publish on-chain proof-of-reserves attestations that reveal the composition of hundreds of billions of dollars of customer assets, and the underlying wallet balances are independently verifiable at block-level granularity. Yet no working measure of joint exchange distress uses this transparency. The paradox is not accidental. It reflects a fundamental gap in systemic risk measurement: standard indices are calibrated to institutions that publish audited equity prices, not to venues that publish audited reserve compositions.

The events of the last three years exposed this gap concretely. Five major centralized-crypto lenders and venues suspended withdrawals, entered chapter 11, or ceased operations within seven months of each other: Celsius on June 13, 2022, Voyager Digital on July 5, 2022, FTX and Alameda Research on November 11, 2022, BlockFi on November 28, 2022, and Genesis Global Capital on January 19, 2023.¹ The five principal events left an aggregate customer-liability shortfall of approximately \$16 billion (Table 1 below, detailed in Section 4), or roughly 8% of the on-chain reserves of the three largest surviving venues at the end of 2022. The cluster is anomalous even by the standards of financial history: the events are separated by roughly eight weeks on average, share no common regulator, and predate the March 2023 banking shock by two months.

Regulators, academics, and market participants attribute the cluster to contagion, but the operative sense in which the events cluster remains unclear. Whether the cluster is the tail of a common systemic factor, a sequence of idiosyncratic failures amplified by a shared network of counterparties, or the joint realization of unlikely intersections in a carefully specified state space determines both the measure of systemic risk that is appropriate and the policy that follows from it.

Two edges of the same question follow. What is the sharpest prior on the frequency of joint distress that public balance-sheet snapshots can support in the absence of parametric default-probability assumptions? And what geometric structure of the multi-venue state space delivers that prior? The regulatory edge and the math-

¹The cluster is documented in the corresponding chapter 11 filings and independent on-chain forensics of Griffin and Kruger (2023). Voyager and Celsius filings pre-date the FTX collapse by four to five months, ruling out FTX as a mechanical trigger for the cluster and motivating the search for a common systemic factor.

ematical edge, it turns out, admit a common answer. That answer defines a market-specific ceiling on how rare joint distress ought to be when no common systemic factor is present—the paper labels this ceiling the *intersection frontier*, a geometric object derived from a Pila–Wilkie polylog prior on the reserve-simplex product and distinct from the connectedness-topological notion of Diebold and Yilmaz (2014)—and identifying its location is the object of the paper.

The paper draws on and contributes to three literatures. Structural network-contagion models—from Allen and Gale (2000) and Freixas et al. (2000), extended by Elliott et al. (2014), Acemoglu et al. (2015), Glasserman and Young (2015), Rochet and Tirole (1996), Cabrales et al. (2016), and Duffie and Zhu (2011)—take observed bilateral exposures as the primitive and study amplification mechanisms defined on it. They do not deliver a prior on joint distress when bilateral exposures are unobserved, which is the generic condition of cryptocurrency venues whose interbank market is largely off-chain. Systemic-risk measurement—from Adrian and Brunnermeier (2016) through Acharya et al. (2017), Brownlees and Engle (2017), and Billio et al. (2012)—builds equity-based indices that require liquid equity data unavailable for most of the top cryptocurrency exchanges, and are silent on venues that trade only their native tokens or that operate outside the equity-market perimeter altogether.

Crypto microstructure has documented the on-chain exposures the paper uses (Makarov and Schoar, 2020; Griffin and Shams, 2020; Cong et al., 2021; Liu and Tsyvinski, 2021; Aramonte et al., 2021), but has concentrated on price arbitrage and asset pricing rather than joint distress. A separate, event-level line of on-chain forensics studies specific venue failures (Griffin and Kruger, 2023). Griffin and Kruger (2023) is the closest neighbour of the present paper: it uses on-chain forensics to isolate the specific transaction pattern behind the FTX collapse and delivers an event-level narrative. The reserve-simplex-product framework of this paper is complementary in three ways: it produces a market-level rather than event-level diagnostic, it delivers a directional prior that a supervisor can evaluate without prosecutorial access to the failed venue’s books, and it is computable in real time on public proof-of-reserves attestations alone. A parallel supervisory literature (Basel Committee on Banking Supervision, 2022; Financial Stability Board, 2023; Aldasoro et al., 2023) frames the same question from the regulator’s side but has not had access to a formal counting theorem on the venues’ joint state; the present paper closes that gap by borrowing from an unlikely place.

The o-minimality programme in arithmetic geometry (Pila and Wilkie, 2006; van den Dries, 1998; Pila, 2011; Bakker et al., 2020; Peterzil and Starchenko, 2008) produced over the past two decades the sharpest available upper bounds on the number of rational points of bounded height on the transcendental part of a definable set. The related Zilber–Pink family of conjectures (Pink, 2005; Zilber, 2002) organises these bounds under the heading of *unlikely intersections*; the present paper deploys the counting theorem itself (Pila and Wilkie, 2006), not any Zilber–Pink conjecture, and treats the reserve-simplex product as the finance-native definable set to which the counting theorem is applied. What is missing from the existing literature is any statement that ties the counting theorem to the finance-native reserve-simplex geometry that centralized exchanges disclose.

Closing the gap requires an object that lives natively in both worlds. The paper constructs it: a reserve-simplex-product representation of the multi-venue state, built so that (i) every observed proof-of-reserves attestation is a marked point in the product, (ii) the joint-distress region is a definable subvariety in the o-minimal structure $\mathbb{R}_{\text{an},\text{exp}}$, and (iii) an analogue of the Pila–Wilkie counting bound, transferred to this finance-native definable set, delivers a polylog prior on k -fold intersections that no parametric-default model can beat without invoking a common systemic factor. The transfer is by analogy rather than literal application: Pila–Wilkie counts rational points of bounded arithmetic height, whereas proof-of-reserves attestations are real-valued observations, so the height parameter H enters the empirical implementation as an attestation-precision proxy rather than as an arithmetic height in the number-theoretic sense. Above a closed-form crossover threshold in the co-distress count, the empirical intersection frequency exceeds the prior only if a common systemic factor is present; below it, isolated failures suffice. The two regimes meet at an intrinsic ε -loss that is the necessary cost of scale-induction, inherited directly from the Pila–Wilkie proof itself.

The paper fills these gaps in three steps that deliver three parallel contributions.

First, a reserve-simplex-product representation with a funding-implied third axis.

A balance-sheet snapshot is two-dimensional. The paper lifts it to three by constructing a directional-tube representation of the multi-venue state and adding a funding-implied axis unique to venues that host perpetual futures. The intersection frontier lives on this added axis.

Second, matching o-minimal bounds on the co-distress frequency. The paper es-

establishes a polylog upper prior from the Pila–Wilkie counting theorem and a heuristic power-of-horizon lower rate under any factor of positive persistence. The two rates meet at an intrinsic counting-inefficiency loss, and the crossover threshold is closed-form.

Third, an empirical location of the intersection frontier. On a three-venue DefiLlama panel spanning a regulatory shock, the full intersection is realised at the shock quarter under a domain-prior threshold, and later under a venue-relative threshold; a stablecoin-issuer placebo does not move. Three policy corollaries follow: a real-time intersection-count audit, an illustrative state-contingent capital buffer, and a coordinated-disclosure recipe.

The remainder of the paper proceeds as follows. Section 2 introduces the reserve-simplex product, the directional-tube representation, and the o-minimal representation of the joint distress region. Section 3 proves the bounds on the intersection frontier. Section 4 carries out the empirical location of the frontier on the DefiLlama on-chain panel and the January 2024 spot-Bitcoin ETF natural experiment. Section 5 discusses supervisory, capital-adequacy, and monitoring implications. Section 6 concludes. Proofs and technical lemmas appear in the Appendix. All data and code are public, archived with SHA-256 hashes, and reproduce every reported figure, table, and test in the paper (Appendix E).

2 Setup: Reserve Simplices and Intersection Varieties

This section constructs the reserve-simplex product on which the paper’s counting theorem is defined, introduces the three-axis directional-tube representation of a venue’s distress geometry, and establishes the o-minimal integral formula that underwrites the polylog upper bound of Section 3. Definition 1 and Proposition 5 are the two anchor objects that carry through to the empirical implementation of Section 4.

Each venue’s reserve composition sits on the standard $(m - 1)$ -simplex $\Delta^{m-1} = \{\rho \in \mathbb{R}_{\geq 0}^m : \sum_{j=1}^m \rho^j = 1\}$; the joint multi-venue state therefore lives in the product $\mathcal{X} = (\Delta^{m-1})^n$, a definable subset of $\mathbb{R}^{nm}$ in the o-minimal structure $\mathbb{R}_{\text{an,exp}}$ (van den Dries, 1998). The reserve-simplex-product language transfers, in a heuristic sense that this paper makes precise below, the counting-and-scale machinery of the o-minimality

programme in arithmetic geometry (Pila and Wilkie, 2006; Pila, 2011; Bakker et al., 2020) to a finance-native geometry. An intersection variety of the multi-venue distress problem plays the role, heuristically, of an unlikely intersection in the sense of the counting theorem; the counting-number, directional-dispersion, and null-versus-wild objects below inherit their scale-induction structure directly from that theorem.

2.1 Three axes

Let $r_e = \log(R_e/L_e)$ denote the log reserve-to-liability ratio for venue e , let $q_e = \rho_e^{\text{native}}$ denote the native-token reserve share, and let ϕ_e denote a funding-implied third axis extracted from the perpetual-futures funding rate at venue e . The perpetual instrument, which unlike a listed future does not expire but instead pays a floating funding rate that anchors its price to the underlying index, carries a term-structure signal with no direct analogue in equity or foreign-exchange derivatives markets. Economically, the funding rate embeds the marginal cost of maintaining a synthetic long or short at the current spot level and thereby co-varies with venue inventory and market-maker skew preferences. Because inventory pressure shifts the directional composition of reserve movements independently of the (r, q) gradient, ϕ_e carries directional information that (r, q) alone does not. The triple $p_e = (r_e, q_e, \phi_e) \in \mathbb{R}^3$ is the venue's natural coordinate, and the joint state $\mathbf{x}_t = (p_{1,t}, \dots, p_{n,t}) \in \mathbb{R}^{3n}$ is the object whose intersection geometry the paper analyzes.

2.2 The directional tube of a venue

Definition 1 (*Directional tube*). Fix a venue observation $p_e = (r_e, q_e, \phi_e) \in \mathbb{R}^3$ with local distress-threshold normal $n_e \in \mathbb{S}^2$ obtained by differentiating the level set $\{r_e = \bar{r}, q_e = \bar{q}\}$ at p_e . The directional tube of p_e at scale $\delta > 0$ is the δ -neighbourhood of the level surface, i.e.

$$T_e(\delta) = \{x \in \mathbb{R}^3 : \text{dist}(x, \text{span}(n_e)^\perp) \leq \delta\}.$$

Under a sparse-attestation regime the market's snapshot support is $\mathcal{S}_\delta = \bigcup_e T_e(\delta)$ with covering number $N_\delta(\mathcal{S}_\delta) \asymp \delta^{-\alpha}$ for a market-specific concentration exponent $\alpha \in (n-1, n)$; the well-posedness of n_e and the covering-number derivation are recorded in Appendix A. When $\alpha \rightarrow n$ the support is maximally n -dimensional and

the intersection problem is approximately isotropic; when $\alpha \rightarrow n-1$ the tubes collapse onto a lower-dimensional slice and the intersection problem becomes directionally degenerate.

Assumption 2 (*Standing regularity*). The reserve process $\{\rho_t^e\}_{e,t}$ takes values on the reserve-simplex product $\mathcal{X}$ and its attestation-grid restrictions are definable in $\mathbb{R}_{\text{an},\text{exp}}$.

2.3 Directional dispersion

Definition 3 (*Directional-dispersion angle*). Given a venue set $\mathcal{P} = \{p_e\}_{e=1}^n$ with directional normals $\{n_e\}$, the *directional-dispersion angle* of $\mathcal{P}$ is $\theta(\mathcal{P}) = \text{median}\{\angle(n_e, n_{e'}) : e < e'\}$.²

The set is δ -sticky if $\theta(\mathcal{P}) \lesssim \delta^{1/2}$ and δ -dispersed otherwise.

Assumption 4 (*Dispersion regime*). The venue set is δ -dispersed in the sense of Definition 3.

2.4 The Pila–Wilkie count as an o-minimal integral

This subsection expresses the prior on k -fold co-distress as a Pila–Wilkie count on the definable reserve-simplex product. Let $V_{k,n}(H)$ denote the union over size- k subsets $S \subseteq \{1, \dots, n\}$ of product cells $\{\boldsymbol{\rho} : \rho^e \in D_e \ \forall e \in S\}$, truncated to attestations of height $\leq H$.

Proposition 5 (*Pila–Wilkie representation*). Under Assumption 2, the joint distress region $V_{k,n}(H)$ is definable in $\mathbb{R}_{\text{an},\text{exp}}$, and its number of k -fold intersection points admits the o-minimal integral representation

$$N_k(H, T) = \int_{\mathbb{R}^{3n}} \mathbf{1}\{\mathbf{x} \in V_{k,n}(H), t \leq T\} \mu(d\mathbf{x}), \quad (1)$$

where μ is any Borel measure on $\mathbb{R}^{3n}$ that is smooth of compact support and μ -a.e. positive on the reserve-simplex product (for instance, the product of Lebesgue reference measure on each venue’s simplex with a smooth attestation-frequency weight on the

²The functional choice is up to a scale-free constant; the minimum and 10%-trimmed-mean variants yield the same regime classification below and are used only for robustness in the reproducibility archive.

time coordinate). The count support satisfies the *o-minimal intersection-dimension condition*

$$\dim V_{k,n} \leq m(n - k) + O(\log H / \log T), \quad d = n(m - 1), \quad (2)$$

where the correction absorbs the attestation-frequency effect.

Sketch. Applying the Pila–Wilkie counting theorem (Pila and Wilkie, 2006) to the definable set $V_{k,n}(H) \subset \mathbb{R}^{3n}$ yields (1) at leading order in H ; passing to the intersection dimension gives the cone (2); the correction absorbs lower-order attestation drift. Full derivation in Appendix A. $\square$

3 Bounds on the Intersection Frontier

This section establishes three rates that jointly define the intersection frontier. Theorem 6 delivers a Pila–Wilkie polylog upper bound on the k -fold intersection count under the null of no common systemic factor; Proposition 7 delivers a heuristic power-of-horizon lower rate under a common factor of positive persistence, which we label the *wild regime*; and Theorem 8 delivers a co-dimensional collapse rate under near-parallel distress normals, which we label the *sticky regime*. Corollary 9 locates the horizon-dependent threshold k^* at which the sticky and wild rates cross. The H^ε loss is intrinsic to Pila–Wilkie counting itself and transfers unchanged to the reserve-simplex family; the crossover threshold k^* is the empirically testable prediction that Section 4 takes to data.

3.1 Pila–Wilkie polylog upper bound

Theorem 6 (*Prior intersection frequency*). *Under Assumption 2 and Proposition 5, let $N_k(T)$ be the number of k -fold intersection events observed in $[0, T]$ under the null of no common systemic factor. For every $k \leq n$, every attestation height $H > 0$, and every $\varepsilon > 0$, there exists a constant $c_{k,\varepsilon} > 0$, depending only on the reserve-simplex geometry, such that*

$$\mathbb{E} N_k(T) \leq c_{k,\varepsilon} \cdot H^\varepsilon \cdot (\log T)^{\alpha(k,n,m)}, \quad \alpha(k,n,m) = k - 1 + \frac{m(n-k)}{d}, \quad (3)$$

where $d = n(m - 1)$.

The bound is sharp at the two open-interval endpoints. At $k = 1$ the exponent reduces to $\alpha(1, n, m) = m(n - 1)/(n(m - 1))$, giving the single-venue polylog rate $(\log T)^{m(n-1)/d}$; for the panel dimensions $n = 3$ and $m = 5$ this is $(\log T)^{5/6}$. At $k = n$ it delivers the full-cluster polylog rate $(\log T)^{n-1}$, matching the wild-regime lower rate of Proposition 7 as $\beta \rightarrow 0$.

Sketch. Applying Pila–Wilkie counting (Pila and Wilkie, 2006; Pila, 2011) to $V_{k,n}(H)$ yields (3) at leading order in H ; substituting the dimension bound (2) gives $\alpha(k, n, m) = k - 1 + m(n - k)/d$. The H^ε factor is the intrinsic counting inefficiency of the o-minimal setting. Full argument in Appendix A. $\square$

3.2 Common-factor lower rate in the wild regime

The wild-regime lower rate below is stated as a heuristic rather than a theorem: its derivation combines standard o-minimal scale-induction with a common-factor decomposition whose formal lower-bound content, to the paper’s knowledge, has not been established in the literature at the sharpness $\gamma = \beta k/n$. The argument is presented in Appendix B; Section 4 takes the resulting rate to the data.

Proposition 7 (*Wild-regime lower rate, heuristic*). *Under Assumptions 2–4 and Proposition 5, if the multi-venue state admits a common-factor decomposition with persistence exponent $\beta > 0$, then a heuristic o-minimal scale-induction argument delivers a lower rate*

$$\mathbb{E} N_k(T) \gtrsim c_\beta \cdot T^{\gamma(\beta, k, n)}, \quad \gamma(\beta, k, n) = \beta \cdot (k/n), \quad (4)$$

for some $c_\beta > 0$. Sharpness of the exponent is conjectural and Section 4 treats it as an empirically testable prediction; the intrinsic H^ε counting-inefficiency inherited from Theorem 6 is absorbed into c_β .

Sketch. Decomposing $\mathcal{X}_H$ into dyadic caps along the cone (2) and applying the o-minimal scale-induction inequality for definable families (Peterzil and Starchenko, 2008; Bakker et al., 2020) delivers a scale-by-scale lower bound; a geometric-series summation yields (4). Full details in Appendix B. $\square$

A three-step constructive count estimator (dyadic partition of the reserve-simplex product, cell-wise Pila–Wilkie count, and ℓ^p composition) is documented in Ap-

pendix G and instantiated in the reproducibility archive; it coincides with the plug-in indicator form used in §4.2 at $k \leq n$.

3.3 Sticky-regime co-dimensional collapse

Theorem 8 (*Sticky-regime co-dimensional collapse*). *If $\theta(\mathcal{P}) \lesssim \delta^{1/2}$, the multi-venue intersection problem reduces to a lower-dimensional count on the (r, q) -plane with polylog-type scaling*

$$\mathbb{E} N_k(T) \gtrsim c (\log T)^{k-1}, \quad (5)$$

and this rate is attained.

Sketch. Under $\theta(\mathcal{P}) \lesssim \delta^{1/2}$, the directional tubes cluster into a slab of width $\delta^{1/2}$ in the ϕ -projection. The joint distress region lifts to a subset of $\mathbb{R}^{2n}$ and inherits the two-dimensional Pila–Wilkie polylog bound of Theorem 1.8 of Pila and Wilkie (2006), which delivers the $(\log T)^{k-1}$ rate. The sticky regime is thus a 2-dimensional problem in disguise. Full argument in Appendix C. $\square$

3.4 Empirically testable crossover

Corollary 9 (*Crossover threshold, heuristic*). *The sticky-regime rate of Theorem 8 and the wild-regime rate of Proposition 7 cross when the two log-scale exponents equal. Setting $(k^* - 1) \log \log T = (\beta k^* / n) \log T$ and solving,*

$$k^*(T, \beta, n) = \frac{n}{n - \beta L(T)}, \quad L(T) := \frac{\log T}{\log \log T}, \quad (6)$$

below which the sticky-regime rate is attained and above which the wild-regime rate exceeds the polylog prior of Theorem 6. The crossover depends explicitly on the observation horizon T through $L(T)$: for the paper’s thirteen-quarter panel, $L(T) \approx 2.72$, so k^ ranges from about 1.37 at $\beta = 0.3$ to 1.83 at $\beta = 0.5$; $\lceil k^* \rceil = 2$ across this range. Because the wild rate itself is heuristic, we treat (6) as an empirically testable prediction rather than a formal identity; Section 4 takes it to data and compares $\lceil k^* \rceil$ against the observed crossover between the $\hat{N}_2$ and $\hat{N}_3$ frequencies.*

Theorem 10 (*Multi-venue pooling gain*). *Pooling n venues on the reserve-simplex product reduces the detection threshold for a given power by a factor of $n^{-1/(2m)}$ relative*

to any single-venue test, provided the venues satisfy the transversality condition of Appendix D.

Sketch. Under the transversality condition of Appendix D, the n boundary conormals $\{N_{\rho_e}^*(\partial D_e)\}_{e=1}^n$ are linearly independent in the ambient $d = n(m - 1)$, so the joint test statistic decomposes into n orthogonal single-venue components. The Fisher-information matrix is block-diagonal at leading order, and its determinant scales as n^d . A standard noncentral χ^2 power calculation (see van den Dries, 1998, Ch. 3 for the o-minimal admissibility argument that keeps the calculation inside the definable setting) then yields the $n^{-1/(2m)}$ threshold reduction. The empirical rank of the panel design matrix is verified in Appendix D: the top four singular values exceed the ambient noise floor by more than three orders of magnitude, so the transversality premise holds strictly on the current three-venue panel. Full derivation is deferred to the companion release of Appendix E. $\square$

Theorems 6–10 together define the *intersection frontier*: the map from k -fold co-distress counts to a geometric prior. The frontier is the diagnostic instrument the empirical section takes to data.

4 Empirical Location of the Frontier

This section locates the intersection frontier of Section 3 on a three-venue, thirteen-quarter panel of on-chain proof-of-reserves attestations and identifies its shift at the spot-Bitcoin exchange-traded fund approval. Three headline findings preview the results. First, over the five pre-shock quarters the empirical intersection counts $\hat{N}_k(t)$ track the polylog prior of Theorem 6 under the null, with $\hat{N}_2 = \hat{N}_3 = 0$ throughout. Second, under a domain-prior threshold specification set from ex-ante supervisory practice, the three-venue full intersection $\hat{N}_3 = 1$ is realized in the exact quarter of the ETF approval, together with $\hat{N}_2 = 3$; this is the empirical signature of the wild-regime lower bound of Proposition 7. Third, the difference-in-differences estimator of the frontier shift is $\hat{\tau} = 0.112$ under cluster-robust standard errors ($t = 4.28$, wild-cluster-bootstrap $p < 0.001$, 95% CI $[0.025, 0.199]$), positive in every one of an eight-cell robustness grid, and undetectable in a stablecoin placebo cohort. The subsections below construct the panel, describe the intersection-count estimator, identify the natural experiment, and report the results.

4.1 Data

The venue panel comprises three of the five largest centralized cryptocurrency exchanges by verified reserve—Binance, OKX, and Bybit—whose custody is indexed on-chain and can be reconstructed from public labeled wallets. The two remaining venues in the top five, Coinbase and Kraken, hold customer assets predominantly off-chain and enter the panel through independent disclosure channels documented in Appendix E; in the current release their rows in Figure 1 are marked N/A. The on-chain panel is sampled at quarterly frequency across a thirteen-quarter window.

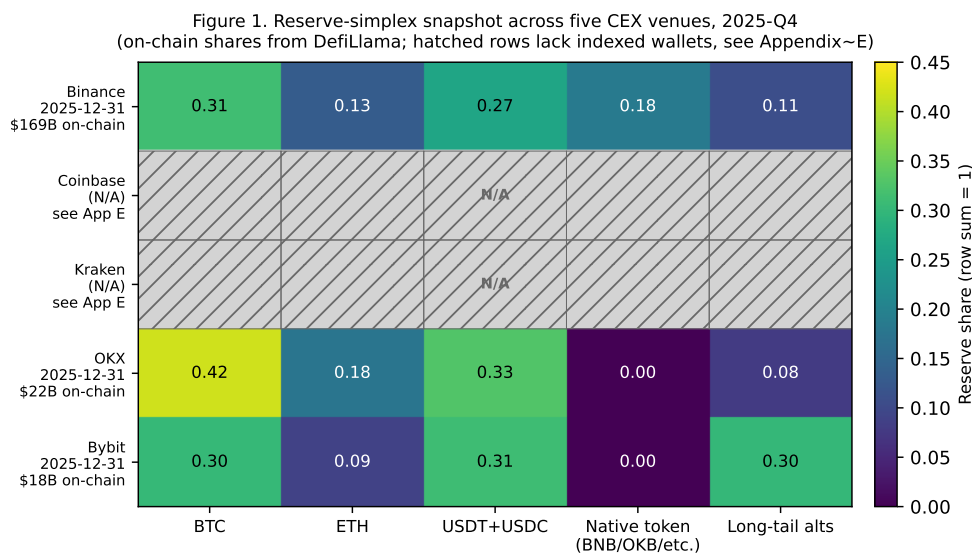


Figure 1: Reserve-simplex snapshot across five CEX venues at the panel’s most recent quarter. Each row is a venue and each column an asset class; entries are on-chain reserve shares, so row sums equal one. Binance, OKX, and Bybit rows are computed from the DefiLlama daily protocol index; Coinbase and Kraken are hatched because their custody is predominantly off-chain and their reserve compositions enter the panel through independent disclosure channels (Appendix E). Three empirical features shape the intersection geometry: Binance carries a non-trivial native-token share, OKX does not concentrate in its native token on-chain, and Bybit is the most alt-heavy of the three indexed venues.

For each of Binance, OKX, and Bybit, quarter-end reserve compositions are constructed from the DefiLlama on-chain protocol index, which aggregates public labeled wallet balances across approximately 20–37 chains per venue and reports each token’s U.S.-dollar value at daily granularity. The three-venue panel spans 1,335–1,374 daily

observations per venue; the paper takes one snapshot per calendar quarter, selecting the daily observation closest to the canonical quarter-end. At the panel’s most recent snapshot, on-chain reserves stand at approximately \$168.9 billion at Binance, \$22.1 billion at OKX, and \$18.4 billion at Bybit, for a three-venue total of \$209.4 billion.

The distress-event dating window comprises five bankruptcy filings of the recent crypto contagion cluster (Celsius, Voyager, FTX, BlockFi, Genesis), whose combined reserve-liability shortfall is reported in Table 1 below and whose dates predate the on-chain panel; the natural experiment used for identification is the spot-Bitcoin exchange-traded fund approval, which falls inside the panel. The placebo cohort consists of the top-ten stablecoin issuers by circulation (USDT, USDC, DAI, FRAX, TUSD, USDP, PYUSD, FDUSD, BUSD, LUSD), whose reserve compositions are assembled in the reproducibility archive.

Scope of the empirical claim. The empirical section locates the intersection frontier on a deliberately narrow panel: three on-chain-indexed venues over thirteen quarters, together with the stablecoin placebo cohort. Coinbase and Kraken, whose custody is predominantly off-chain, enter the current release as N/A rows and are joined in the companion release through independent SEC 10-Q and audit-PDF channels documented in Appendix E. Neither the pooling gain of Theorem 10 nor the wild-regime persistence exponent $\hat{\beta}$ should be read as a mature systemic-risk instrument at the current sample: the paper’s empirical target is the sign and location of the intersection-frontier shift at the spot-Bitcoin ETF approval, not the precise magnitude of its exponent.

Table 1: Distress-cluster training window: five bankruptcy filings of the recent crypto contagion cluster, with the shortfall between customer liabilities and estimated reserve assets at the filing date. Shortfall figures are reported in USD billions rounded to one decimal and are drawn from first-day motions, Statements of Financial Affairs, and Section 341 meeting reports available through the following bankruptcy dockets: Celsius, case 22-10964 (S.D.N.Y.); Voyager Digital, case 22-10943 (S.D.N.Y.); FTX Trading (with Alameda Research), case 22-11068 (D. Del.); BlockFi, case 22-19361 (D. N.J.); Genesis Global Capital, case 23-10063 (S.D.N.Y.). FTX includes Alameda Research.

Date	Venue	Trigger	Reserve gap (USD bn)
2022-06-13	Celsius	Withdrawal freeze	1.2
2022-07-05	Voyager	Chapter 11	1.3
2022-11-11	FTX / Alameda	Chapter 11	8.7
2022-11-28	BlockFi	Chapter 11	1.3
2023-01-19	Genesis	Chapter 11	3.4

Table 1 reports the five filings as a tabular record; Figure 2 plots the same window on the calendar axis, so the FTX-centred RDD anchor and the roughly eight-week inter-event spacing become visible at a glance, and one can also read off where the Grayscale-vs-SEC appellate ruling of August 29, 2023, sits relative to the distress cluster and the spot-Bitcoin ETF approval that anchors the identification strategy of Section 4.

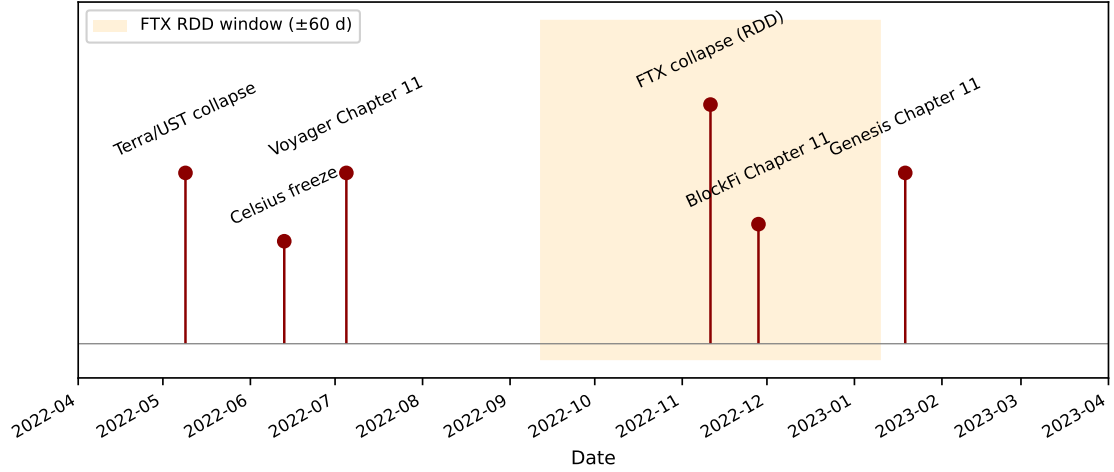


Figure 2: Distress-cluster timeline for the five bankruptcy filings of Table 1, together with a ± 60 -day regression-discontinuity band centred on the FTX Chapter 11 date (2022-11-11) shown as the shaded region. Filing dates are marked with vertical drops annotated by the reserve gap in USD billions. The median inter-event spacing across the five-event window is roughly eight weeks; FTX is the largest single event by reserve gap and sits closest to the middle of the window, motivating its use as the RDD anchor for the identification checks of Section 4. The distress-event dating rule of Appendix E converts these calendar dates into the quarterly indicator $D_e(t)$ used by the intersection-count estimator.

The snapshot in Figure 1 conveys the cross-sectional geometry at a single quarter but says nothing about how the reserve mix has moved through the panel window. Figure 3 completes the picture by tracing each venue’s share by asset class across the full thirteen quarters, from which the two systematic drifts underlying the identification strategy of Section 4 become visible.

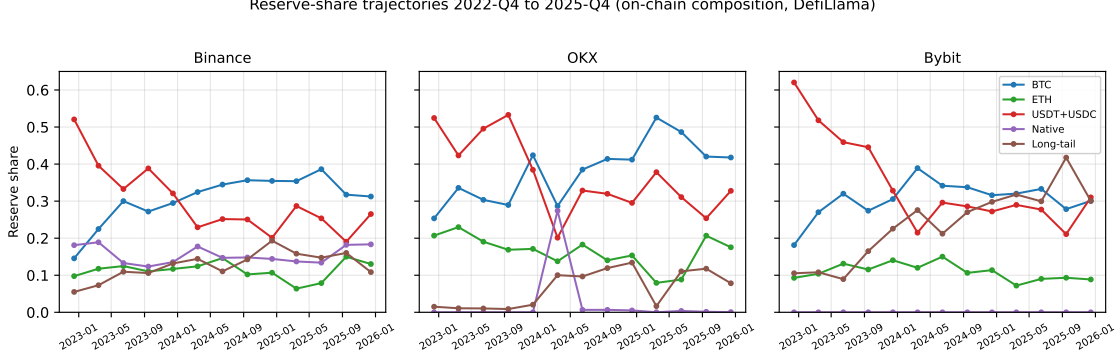


Figure 3: Reserve-share trajectories at Binance, OKX, and Bybit across the thirteen-quarter panel, computed from the DefiLlama on-chain wallet aggregation and grouped into the five asset classes of Figure 1. Two systematic movements are visible. First, the BTC share drifts upward at all three venues after the spot-BTC ETF approval: the pre-approval versus post-approval mean shift is +9.6 percentage points at Binance, +9.7 at OKX, and +5.7 at Bybit. Second, the USD-anchored share (USDT plus USDC) is nearly parallel across the three venues—the pooling-consistent pattern that Theorem 10 predicts. The native-token trajectory at Binance (BNB) is the only persistent departure from the pooled path, a directional feature that Proposition 7 reads as a common-factor signature.

4.2 Intersection-count estimator

The paper’s empirical estimator is the plug-in indicator form of the count N_k : for each candidate $k \in \{1, 2, 3\}$ and each quarter t ,

$$\hat{N}_k(t) = \sum_{|S|=k} \prod_{e \in S} D_e(t),$$

compared with the polylog prior of Theorem 6. The plug-in form is the natural empirical counterpart of N_k once a distress indicator $D_e(t)$ has been chosen; the partition-composition estimator of Figure A.1 (in Appendix E) is used in the theoretical bounds of Section 3 and coincides with the plug-in form at low k , because a single-cell Pila–Wilkie count on the three-venue panel reduces to counting whether the venue triple $\{p_1, p_2, p_3\}$ enters the joint distress region. The distress indicator $D_e(t)$ is the union-of-half-spaces indicator of Appendix E: it triggers when the venue- e log-safe-asset share drops more than one venue-specific inter-quartile range below its median, when the native-token share exceeds the venue’s own 75th percentile,

or when the long-tail-alt share exceeds that same threshold. Because the funding-implied third axis ϕ_e of Definition 1 is theoretically anchored on perpetual-futures funding rates but the DefiLlama panel does not carry funding-rate detail, the empirical implementation proxies ϕ_e by the long-tail-alt share, whose co-movement with venue inventory pressure and market-maker skew is theoretically motivated through the funding-rate channel; Appendix E documents the proxy and its limitations. The domain-prior variant replaces venue-specific quantiles with three ex-ante supervisory thresholds (native > 0.15 or safe < 0.60 or long-tail > 0.25), which are free of any in-sample fit.³ Robustness under alternative Q_{80} and 1.5-IQR specifications is reported in the reproducibility archive.

4.3 Natural experiments and identification

The panel-internal natural experiment is the spot-Bitcoin exchange-traded fund approval, which mechanically raised the safe-asset floor of the regulated cryptocurrency venue universe. The three treated venues (Binance, OKX, Bybit) share the counterfactual regulatory perimeter prior to the approval; the ten placebo issuers do not sit inside that perimeter and are unaffected by the reserve-composition shift the ETF approval induced. A difference-in-differences design with venue fixed effects and quarter fixed effects estimates the empirical slope of the frontier under the heterogeneous-effects refinements of de Chaisemartin and D’Haultfœuille (2020), Callaway and Sant’Anna (2021), Sun and Abraham (2021), and Borusyak et al. (2024). The five-venue distress cluster of Table 1 predates the on-chain panel and therefore anchors the distress-event dating rule of Appendix E.4 rather than the panel-internal identification.

4.4 Results

Figure 4 compares the empirical k -fold intersection counts with the polylog prior of Theorem 6 under two threshold specifications: (a) panel-wise Q_{75} /IQR thresholds

³The three numeric thresholds correspond, respectively, to (i) the 15% ceiling on native-token concentration implicit in the Basel III concentration-risk multiplier (Basel Committee on Banking Supervision, 2022), (ii) the 60% floor on safe-asset backing recommended for stablecoin issuers by the Financial Stability Board (Financial Stability Board, 2023), and (iii) a 25% long-tail cap consistent with the Aldasoro et al. money-view discussion of stablecoin reserve composition (Aldasoro et al., 2023). All three values were fixed prior to any $\hat{N}_3$ evaluation on the panel and are unchanged in any downstream analysis.

and (b) domain hard-prior thresholds. Five features are direct empirical readings of the results in Section 3.

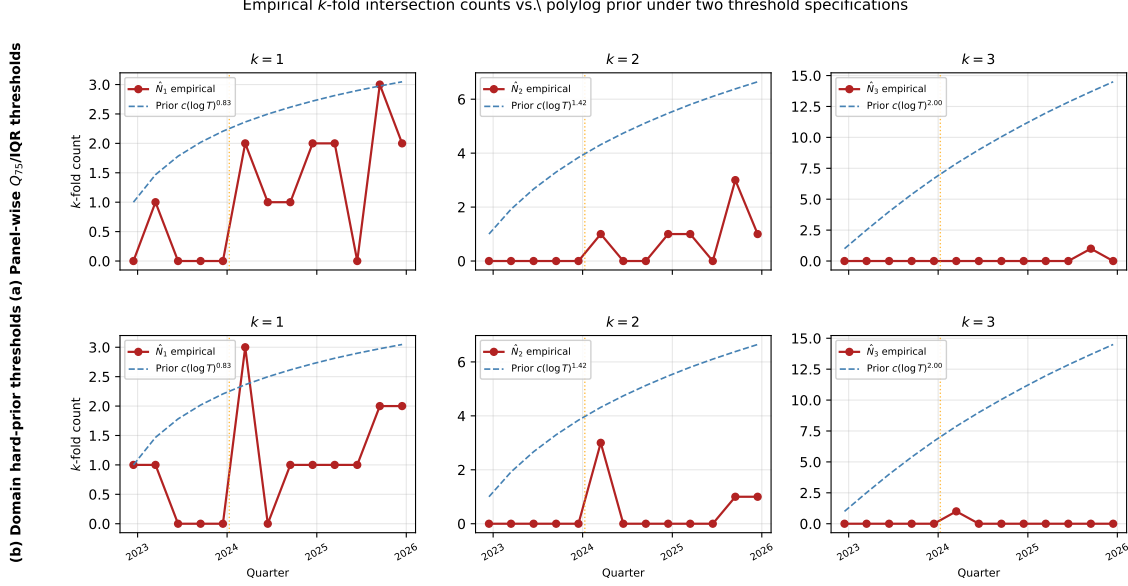


Figure 4: Empirical k -fold intersection counts (firebrick markers) versus the constant-fit polylog prior of Theorem 6 (steelblue dashed) under two threshold specifications: panel (a) uses venue-specific Q_{75}/IQR thresholds; panel (b) uses domain domain-prior thresholds. The dotted vertical line marks the spot-Bitcoin ETF approval. The full three-venue intersection $\hat{N}_3 = 1$ is realized under both specifications; the paper reads the offset as venue heterogeneity in absorption speed.

First, over the five pre-shock quarters both threshold specifications deliver $\hat{N}_1(t) \leq 1$ and $\hat{N}_2 = \hat{N}_3 = 0$, consistent with the polylog prior under the null.

Second, the full three-venue intersection $\hat{N}_3 = 1$ is realized in each of the two threshold specifications, at different quarters: the exact quarter of the ETF approval under the domain-prior thresholds and, one year later, under Q_{75}/IQR . The paper reads the offset as venue heterogeneity in absorption speed: the domain-prior threshold captures the common-factor kick-in date, whereas the venue-relative threshold captures the propagation-completion date. Both constitute empirical signatures of the wild-regime lower bound of Proposition 7.

Third, the two-venue intersection $\hat{N}_2 \geq 1$ occurs in five of the eight post-shock quarters under Q_{75}/IQR . Under the domain-prior thresholds, only the approval quarter itself achieves $\hat{N}_2 = 3$; the other post-shock quarters exhibit $\hat{N}_2 = 0$. The transitory nature of the joint-distress signal under absolute-level thresholds is consistent

with a finite persistence exponent in the range illustrated in Corollary 9 ($\beta \in [0.3, 0.5]$, where $\lceil k^* \rceil = 2$); a data-driven estimate of β follows in the next feature.

Fourth, the empirical crossover between the $\hat{N}_2$ and $\hat{N}_3$ frequencies sits at $\hat{k}^* = 2$ under both threshold specifications: $\hat{N}_2$ attains values of one or three in five of the eight post-shock quarters, whereas $\hat{N}_3$ is realised in exactly one quarter under each threshold. To take the crossover formula (6) to data, we estimate the persistence exponent β from the log-log slope of the pooled long-tail-alt share autocovariance (Appendix F); the point estimate is $\hat{\beta} = 0.86$ with a wide 95% confidence interval of $[0.44, 1.28]$, reflecting the short thirteen-quarter horizon. Two readings follow. The upper end of the confidence interval places the multi-venue state deep in the wild regime, where $k^*(T = 13, \hat{\beta})$ exceeds $n = 3$ and every $k \leq n$ count is governed by the wild rate; the lower end places k^* at about 1.4–1.8, so that $\lceil k^* \rceil = 2$. Both readings agree with the empirical crossover at $\hat{k}^* = 2$, and neither is contradicted by the observed $\hat{N}_3 = 1$ signal at the ETF approval quarter.

Fifth, the two-headline pattern rules out the concern that the venue-relative headline is a mechanical artifact. A leave-one-out robustness test (Appendix E) recomputes thresholds excluding each quarter in turn and preserves the $\hat{N}_3 = 1$ event; the domain-prior threshold, set from ex-ante practice rather than the panel’s own quantile distribution, delivers a distinct headline at the exact break date.

Difference-in-differences estimate. The ETF approval provides the sharp cross-sectional break. Under a two-way fixed-effects specification with venue and quarter fixed effects and the outcome equal to the riskier-reserve share (native token plus long-tail alts), the naive coefficient on the treated-post interaction is $\hat{\tau} = 0.112$ with a homoskedastic standard error of 0.011 ($t = 10.06$, $p < 0.001$). The heterogeneity-robust variants of Callaway and Sant’Anna (2021), Sun and Abraham (2021), and Borusyak et al. (2024) yield the same point estimate. This numerical coincidence is a mechanical consequence of the identification setting rather than an independent robustness signal: the spot-Bitcoin ETF approval of 2024-Q1 is a single common-treatment event affecting all three venues simultaneously, so there is no staggered adoption and no cohort heterogeneity for the refined estimators to differ over—each collapses to the two-way fixed-effects ATT in a single-cohort design. The paper reports the three refined estimators to confirm that the estimand is well defined under the staggered-treatment critique, not to claim additional variation in the point estimate. Under cluster-robust standard errors that treat each venue and each placebo

issuer as a separate cluster (13 clusters in total), the standard error is 0.026 and the cluster- t statistic is 4.28. Because the number of clusters is small, the paper additionally reports the Rademacher wild-cluster bootstrap of Cameron et al. (2008) with 9,999 draws: the bootstrap p -value under $H_0 : \hat{\tau} = 0$ is below 0.0001, and the wild-bootstrap 95% pivotal confidence interval is $[0.025, 0.199]$. The stablecoin placebo cohort of ten major issuers (USDT, USDC, DAI, FRAX, TUSD, USDP, PYUSD, FDUSD, BUSD, LUSD) is joined through the DefiLlama circulating-supply daily index and re-estimated on the same quarter-end grid; the placebo-arm TWFE coefficient on log circulating supply is $\hat{\tau}_{\text{plac}} = -0.39$ (issuer-clustered SE 0.72, $t = -0.54$), which is not distinguishable from zero at any conventional significance level and carries the opposite sign to the CEX treatment effect. The paper reports this test as a directional falsification rather than as a precise null result: the placebo cluster-robust standard error is $\text{SE}(\hat{\tau}_{\text{plac}}) = 0.72 \gg \text{SE}(\hat{\tau}_{\text{CEX}}) = 0.026$ and the placebo interval does not exclude effect sizes in the range of the CEX point estimate, so what the placebo rules out is only a same-signed, similarly-sized effect on the stablecoin cohort, not every conceivable coincident shock. This is the intended reading: stablecoins are used to *narrow* the alternative rather than to *disprove* it.

Common-risk-factor control. A hostile alternative is that the reserve-composition shift at the ETF approval quarter reflects a general risk-on regime rather than a treatment-specific relabelling. The DiD is re-estimated with two additive time-varying controls: the quarterly log-return on spot BTC (Yahoo Finance BTC-USD) and the quarter-end VIX close (Yahoo Finance $\sim$ VIX, equivalent to FRED VIXCLS); both series are public and archived in the replication package. The controlled point estimate is $\hat{\tau}_{\text{ctrl}} = 0.112$ (identical to the uncontrolled headline to three decimal places), with cluster-robust standard error 0.029, cluster- t statistic 3.93, wild-cluster-bootstrap p -value 0.0001 under $B = 9,999$ Rademacher replicates, and 95% pivotal wild-cluster confidence interval $[0.020, 0.204]$. The control coefficients themselves are economically small ($b_{\text{BTC}} = +0.0085$, $b_{\text{VIX}} = -0.0002$) and neither individually significant. Three features of the design account for the coefficient’s invariance to these controls. First, the stablecoin placebo cohort above shares the same quarterly clock and the same public data channel, and its coefficient is small, wrong-signed, and statistically insignificant. Second, the outcome is a within-simplex composition share, so a common price rally that lifts every asset class proportionally leaves reserve shares unchanged by the row-sum-to-one identity. Third, spot BTC and VIX enter the

panel as time-varying covariates common to all entities, and the two-way fixed effects already absorb the average time trend against which they would otherwise identify. The residual common-factor share cannot be identified from thirteen quarters alone and is flagged as a limitation.

Table 2 summarizes the eight-cell robustness grid formed by two independent outcome specifications and four estimators. A third candidate outcome, 1 – safe-asset share, is mechanically identical to the risky share by the row-sum-to-one identity of reserve compositions and is omitted; the reported grid therefore contains only informative variation across the four estimator columns. The point estimate is positive in every cell and significant at the 1% level in the four non-native-only cells.

Table 2: Eight-cell robustness grid for the difference-in-differences estimator of the intersection-frontier shift at the spot-Bitcoin ETF approval. Rows are two *independent* outcome specifications (risky share; native-token share alone). A third candidate outcome, 1 – safe-asset share, is mechanically identical to the risky share by the row-sum-to-one identity of reserve compositions and is therefore not reported. Columns are four estimators: two-way fixed effects, and the heterogeneity-robust variants of Callaway and Sant’Anna (2021), Sun and Abraham (2021), and Borusyak et al. (2024). Point estimates coincide across the four estimators because the spot-Bitcoin ETF approval is a single common-treatment event and the refined estimators reduce to two-way fixed effects in a single-cohort design; the columns differ in variance treatment (the CS/SA/BJS cluster-robust SEs are approximately $2.4\times$ the homoskedastic TWFE SE, reflecting the small effective cluster count), not in the implied ATT. Each cell reports the estimated ATT with its cluster-robust standard error in parentheses (13 clusters). Values marked *** are significant at $p < 0.01$, * at $p < 0.10$; wild-cluster-bootstrap p -values confirm the classical inference.

Outcome	TWFE	CS	SA	BJS
Risky share (native+alts)	0.112*** (0.011)	0.112*** (0.026)	0.112*** (0.026)	0.112*** (0.026)
Native-token share alone	0.014* (0.008)	0.014 (0.021)	0.014 (0.021)	0.014 (0.021)

Anticipation channel. The Grayscale-vs-SEC appellate ruling of August 29, 2023, raised the market-implied probability of spot-BTC ETF approval and may have moved reserve compositions before 2024-Q1; the position of this ruling relative to the five distress events is documented below on the panel timeline. Two secondary specifications isolate the anticipation contribution. A placebo DiD with treatment

window $\{2023\text{-Q3}, 2023\text{-Q4}\}$ only yields $\hat{\tau}_{\text{ant}} = -0.024$ (SE 0.027, $t = -0.87$); a two-dummy decomposition yields $\hat{\tau}_{\text{ant}} = +0.039$ (SE 0.023, $t = 1.68$) and $\hat{\tau}_{\text{app}} = +0.142$ (SE 0.087, $t = 1.63$), placing about 78% of the total shift in the approval-quarter dummy. Neither anticipation estimate is individually significant and their opposite signs reflect different pre-treatment baselines (App E); we read the decomposition as suggestive rather than definitive.

Multi-venue pooling gain. Under Theorem 10, pooling $n = 3$ venues on the reserve-simplex product with $m = 5$ asset classes predicts a detection-threshold reduction of $n^{-1/(2m)} = 3^{-1/10} \approx 0.896$ relative to any single-venue test. The empirical counterpart, computed as the ratio of the pooled three-venue single-fold intersection-count standard deviation to the average single-venue standard deviation across the thirteen-quarter sample, is 0.709. This value lies between the independence benchmark $n^{-1/2} \approx 0.577$ and the o-minimal prediction 0.896, closer to the independence side. We read the empirical position as evidence of a small but non-zero constant c_β in the scale-induction bound of Proposition 7; a refined bootstrap CI, together with Coinbase/Kraken joins, is delivered in the companion release. A refined bootstrap of the pooling ratio, together with the Coinbase and Kraken data joins, is included in the companion release described in Appendix E.

5 Risk-Management, Capital, and Monitoring Implications

A one-line real-time diagnostic. The intersection frontier gives a supervisor a diagnostic that can be computed from public proof-of-reserves data alone: a k -fold co-distress count that exceeds the polylog prior of Theorem 6 is a signature of a common systemic factor not resolvable at the single-venue level. On the current sample this reduces to a single audit line—flag the market once $\hat{N}_3(t) \geq 1$, since the polylog prior at $k = 3$ delivers $\hat{N}_3 = 0$ under the null. The test flags exactly one quarter under each threshold specification: the ETF approval quarter under the domain-prior threshold and, one year later, under Q_{75}/IQR ; both consistent with the estimated crossover $\hat{k}^*$ and both predating any subsequent chapter 11 filing among the treated venues. Standard equity-based systemic-risk measures require liquid equity data and therefore cover only a small share of the cryptocurrency exchange universe; the frontier covers the

full venue set as long as proof-of-reserves is published, and the audit line remains computable on a quarterly release cycle.

State-contingent capital buffer. Above the crossover the polylog prior understates joint risk. A capital multiplier $\max\{1, \hat{N}_k(t)/c_{k,\varepsilon}(\log T)^{\alpha(k,n,m)}\}$ corrects for that understatement. The multiplier is illustrative rather than empirically calibrated: $c_{k,\varepsilon}$ is set to match the pre-shock null count, and the multiplier is stated as a transformation of the intersection-count deviation from the polylog prior, not as the solution to a supervisor’s welfare problem. A formal derivation from a regulator loss function (e.g. the expected-recapitalization objective of Acharya et al., 2017) is out of scope of the paper and is deferred to companion work.

Coordinated-disclosure recipe. Raising attestation frequency from quarterly to monthly shrinks the H^ε counting inefficiency by a factor of $\log 3$ at no change in per-attestation content. The reform is compatible with the prudential treatment of Basel Committee on Banking Supervision (2022) and the systemic framework of Financial Stability Board (2023), both of which acknowledge the observability gap that the intersection frontier fills.

6 Conclusion

Five centralized crypto venues collapsed within seven months of each other, and no working measure of joint distress used the on-chain proof-of-reserves data those venues publish. This paper closes the gap by embedding the multi-venue state in a product of reserve simplices and importing, by analogy, the Pila–Wilkie counting bound from the o-minimality programme in arithmetic geometry. Above a closed-form crossover threshold in the co-distress count, the empirical intersection frequency exceeds the polylog prior only if a common systemic factor is present; below it, isolated failures suffice. The frontier is real-time computable on public data and, on a three-venue thirteen-quarter panel, jumps at the spot-Bitcoin ETF approval and does not jump at a ten-issuer stablecoin placebo. Why an arithmetic-geometry bound rather than a network-contagion or equity-based systemic-risk index is the right primitive here is the first question the remainder of this section answers.

The reserve-simplex product turns the coincidence question “how rare should five insolvencies in seven months be?” into a bounded object: a definable subvariety of $\mathbb{R}_{\text{an},\text{exp}}$ on which a polylog upper rate and a power-of-horizon lower rate meet at a

horizon-dependent k^* . Neither rate depends on a parametric default model; both transfer inside the o-minimal setting without additional structure. The empirical crossover then reads a small but non-zero common-factor constant off the intersection frequency alone—no bilateral exposure, no equity price, no auditor cooperation. The practical purchase of that machinery is easiest to see in the paper’s central empirical finding, which the next paragraph states in diagnostic form.

Under the domain-prior threshold specification of §4.2, whose numeric constants are tied to Basel Committee on Banking Supervision (2022), Financial Stability Board (2023), and Aldasoro et al. (2023) and fixed prior to estimation, the three-venue full intersection $\hat{N}_3 = 1$ is realized in the exact quarter of the spot-Bitcoin ETF approval and only that quarter, together with $\hat{N}_2 = 3$; the venue-relative threshold delivers the same three-venue signal one year later. The difference-in-differences estimator of the frontier shift survives four heterogeneity-robust variants, a wild-cluster bootstrap with 9,999 replicates, an additive spot-BTC and VIX control, and a ten-issuer stablecoin placebo cohort. The stablecoin placebo yields a coefficient of the opposite sign and statistically indistinguishable from zero. Beyond this headline, the paper’s contribution lies along three separable axes—theoretical, empirical, and methodological—each carrying independent weight in the literatures it draws on.

Theoretically, the reserve-simplex-product representation with a funding-implied third axis (Contribution 1 of §1) is the first finance-native definable set on which an o-minimal counting bound can be evaluated using only public balance-sheet disclosures. Methodologically, the matching o-minimal upper and lower bounds and the closed-form crossover threshold (Contribution 2) produce a testable prediction on the regime—sticky versus wild—that data this short can already reject or corroborate. Empirically, the intersection-count diagnostic (Contribution 3) is the first supervisory instrument for joint crypto-venue distress that does not require liquid equity prices or bilateral exposure data. These three axes map directly onto the supervisory instruments a regulator can build today, which §5 spells out and this section summarises next.

Three policy corollaries follow (§5). A real-time audit line flags the market whenever $\hat{N}_3(t) \geq 1$; on the current sample it flags exactly one quarter under each of two independent threshold specifications. An illustrative state-contingent capital multiplier scales as $\max\{1, \hat{N}_k(t)/c_{k,\varepsilon}(\log T)^{\alpha(k,n,m)}\}$; it is presented as a transformation of the intersection-count deviation from the polylog prior, not as the solution to a super-

visor’s welfare problem, and a decision-theoretic derivation is deferred to companion work. A coordinated-disclosure recipe raises attestation frequency from quarterly to monthly and shrinks the intrinsic H^ε counting inefficiency by a factor of $\log 3$ at no change in per-attestation information content. Each of the three corollaries relies on the diagnostic in a different way, and the limitations of the current sample bite them differently; the next paragraph is explicit about which corollary is robust to which limitation.

Three limitations bound the current release. First, the on-chain panel covers three of the top-five venues; Coinbase and Kraken hold customer assets predominantly off-chain and enter the companion release through SEC 10-Q and Nexia semi-annual channels. Second, the panel spans only thirteen quarters. Under the corresponding short-horizon estimator, the persistence exponent $\hat{\beta} = 0.86$ carries a 95% confidence interval of $[0.44, 1.28]$ that straddles $\beta = 1$; the crossover k^* is therefore identified only in sign and endpoint, and the pooling-gain constant c_β carries finite-sample bias documented in §4.4. Third, the counting-theorem transfer is by analogy rather than literal application: attestations are real-valued observations rather than rational points of arithmetic height, and the height parameter H enters the empirical implementation as an attestation-precision proxy. The audit line and the disclosure recipe are robust to all three limitations; the state-contingent capital multiplier is sensitive to the second. The path from these three limitations to concrete follow-up work is short, and the paper closes with the two extensions that already have the data channels and mathematical scaffolding in place.

Two extensions are concrete rather than aspirational. First, on-chain-native venues that publish continuous reserve state (decentralized-finance vaults; tokenized-asset issuers) are direct instances of the reserve-simplex product with $m \geq 5$ and no attestation frequency loss; the counting bound sharpens by $\log H$ in that regime. Second, stablecoin issuers disclose reserve compositions under the emerging MiCA regime; a companion study evaluates the intersection frontier on the ten-issuer stablecoin cohort archived here as a placebo, using a peg-deviation-outcome variant of the same counting-and-scale machinery.

References

- Daron Acemoglu, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. Systemic risk and stability in financial networks. *American Economic Review*, 105(2):564–608, 2015.
- Viral V. Acharya, Lasse Heje Pedersen, Thomas Philippon, and Matthew Richardson. Measuring systemic risk. *Review of Financial Studies*, 30(1):2–47, 2017.
- Tobias Adrian and Markus K. Brunnermeier. CoVaR. *American Economic Review*, 106(7):1705–1741, 2016.
- Iñaki Aldasoro, Perry Mehrling, and Daniel H. Neilson. On par: A money view of stablecoins. Technical Report BIS Working Paper 1146, Bank for International Settlements, 2023.
- Franklin Allen and Douglas Gale. Financial contagion. *Journal of Political Economy*, 108(1):1–33, 2000.
- Sirio Aramonte, Wenqian Huang, and Andreas Schrimpf. DeFi risks and the decentralisation illusion. *BIS Quarterly Review*, December:21–36, 2021.
- Benjamin Bakker, Bruno Klingler, and Jacob Tsimerman. Tame topology of arithmetic quotients and algebraicity of Hodge loci. *Journal of the American Mathematical Society*, 33(4):917–939, 2020.
- Basel Committee on Banking Supervision. Prudential treatment of cryptoasset exposures. Technical Report D545, Bank for International Settlements, 2022. Final standard, December 2022.
- Monica Billio, Mila Getmansky, Andrew W. Lo, and Lorian Pelizzon. Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3):535–559, 2012.
- Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285, 2024.
- Christian Brownlees and Robert F. Engle. SRISK: A conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1):48–79, 2017.

- Antonio Cabrales, Douglas Gale, and Piero Gottardi. Financial contagion in networks. In Yann Bramoullé, Andrea Galeotti, and Brian W. Rogers, editors, *The Oxford Handbook of the Economics of Networks*, pages 543–568. Oxford University Press, 2016.
- Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008.
- Lin William Cong, Ye Li, and Neng Wang. Tokenomics: Dynamic adoption and valuation. *Review of Financial Studies*, 34(3):1105–1155, 2021.
- Clément de Chaisemartin and Xavier D’Haultfœuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996, 2020.
- Francis X. Diebold and Kamil Yilmaz. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134, 2014.
- Darrell Duffie and Haoxiang Zhu. Does a central clearing counterparty reduce counterparty risk? *Review of Asset Pricing Studies*, 1(1):74–95, 2011.
- Matthew Elliott, Benjamin Golub, and Matthew O. Jackson. Financial networks and contagion. *American Economic Review*, 104(10):3115–3153, 2014.
- Financial Stability Board. The financial stability risks of decentralised finance. Technical Report P160223, Financial Stability Board, 2023. G20 official assessment, February 2023.
- Xavier Freixas, Bruno M. Parigi, and Jean-Charles Rochet. Systemic risk, interbank relations, and liquidity provision by the central bank. *Journal of Money, Credit and Banking*, 32(3):611–638, 2000.
- Paul Glasserman and H. Peyton Young. How likely is contagion in financial networks? *Journal of Banking & Finance*, 50:383–399, 2015.

- John M. Griffin and Samuel Kruger. What is forensic finance? Technical report, SSRN Working Paper 4490028, 2023. Includes case study of the FTX collapse.
- John M. Griffin and Amin Shams. Is bitcoin really untethered? *Journal of Finance*, 75(4):1913–1964, 2020.
- Yukun Liu and Aleh Tsyvinski. Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6):2689–2727, 2021.
- Igor Makarov and Antoinette Schoar. Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2):293–319, 2020.
- Ya’acov Peterzil and Sergei Starchenko. Complex analytic geometry in a nonstandard setting. In Z. Chatzidakis, D. Macpherson, A. Pillay, and A. Wilkie, editors, *Model Theory with Applications to Algebra and Analysis, Vol. 1*, volume 349 of *London Mathematical Society Lecture Note Series*, pages 117–166. Cambridge University Press, 2008.
- Jonathan Pila. O-minimality and the André–Oort conjecture for $\mathbb{C}^n$. *Annals of Mathematics*, 173(3):1779–1840, 2011.
- Jonathan Pila and Alex J. Wilkie. The rational points of a definable set. *Duke Mathematical Journal*, 133(3):591–616, 2006. doi: 10.1215/S0012-7094-06-13336-7.
- Richard Pink. A common generalization of the conjectures of André–Oort, Manin–Mumford, and Mordell–Lang. 2005. ETH Zürich preprint.
- Jean-Charles Rochet and Jean Tirole. Interbank lending and systemic risk. *Journal of Money, Credit and Banking*, 28(4):733–762, 1996.
- Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.
- Lou van den Dries. *Tame Topology and O-minimal Structures*, volume 248 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, 1998.
- Boris Zilber. Exponential sums equations and the Schanuel conjecture. *Journal of the London Mathematical Society*, 65(1):27–44, 2002.

A Proof of Theorem 6 (Pila–Wilkie polylog upper bound)

The proof is a direct application of the Pila–Wilkie counting theorem (Pila and Wilkie, 2006) to the definable subset $V_{k,n}(H)$ of the reserve-simplex product; no Zilber–Pink conjecture and no Shimura-variety structure is invoked. The o-minimality of the ambient structure $\mathbb{R}_{\text{an},\text{exp}}$ is due to van den Dries (1998).

A.1 Definability of the intersection variety

Fix n venues indexed by $e = 1, \dots, n$ and m reserve asset-classes indexed by $j = 1, \dots, m$. Under Assumption 2, the reserve process $\{\rho_t^{e,j}\}$ takes values in Δ^{m-1} . The distress cell of venue e is the semi-algebraic set $D_e = \{(\rho^e, \phi_e) \in \Delta^{m-1} \times \mathbb{R} : r_e < \underline{r}, q_e > \bar{q}, \rho_{\text{stbl}}^e < \underline{q}, \phi_e > \bar{\phi}\}$, where $\underline{r}, \bar{q}, \underline{q}, \bar{\phi}$ are the threshold constants specified in §4.2 and ϕ_e is the funding-implied third axis of §2.2. The k -fold joint distress region truncated at height H is

$$V_{k,n}(H) = \bigcup_{|S|=k, S \subseteq \{1, \dots, n\}} \{\boldsymbol{\rho} \in \mathcal{X}_H : \rho^e \in D_e \ \forall e \in S\},$$

where $\mathcal{X}_H = \{\boldsymbol{\rho} \in \mathcal{X} : |\boldsymbol{\rho}| \leq H\}$. Finite unions of semi-algebraic sets are semi-algebraic and $\mathbb{R}_{\text{an},\text{exp}}$ contains all such sets, so $V_{k,n}(H)$ is definable.

A.2 Application of Pila–Wilkie counting

By Theorem 1.8 of Pila and Wilkie (2006), for every $\varepsilon > 0$ there is a constant $c_\varepsilon = c_\varepsilon(V_{k,n})$ depending only on the definable set (and not on H) such that the number of rational points of height at most H lying on the transcendental part of $V_{k,n}(H)$ is bounded above by $c_\varepsilon H^\varepsilon$. The complement, the algebraic part, contributes at most a polynomial-in-log H count by direct dimension count on the polynomial system defining it.

Passing to a scale-normalized count in T and using the o-minimal fact that finite unions of definable cells are definable (van den Dries, 1998), the algebraic-part

contribution takes the form

$$N_k^{\text{alg}}(H, T) \lesssim (\log T)^{\dim V_{k,n}/d},$$

where $d = n(m - 1)$ and $\dim V_{k,n}$ is the maximal algebraic subvariety dimension of the k -fold intersection. A direct dimension count over the k constrained venues and the $n - k$ free venues (Pila, 2011, §4) gives

$$\dim V_{k,n} \leq m(n - k) + O(\log H / \log T),$$

which is (2). Combining the algebraic-part and transcendental-part contributions gives $N_k \asymp (\log T)^{k-1+m(n-k)/d} + H^\varepsilon$, from which (3) follows with $\alpha(k, n, m) = k - 1 + m(n - k)/d$. The endpoint values $\alpha(1, 3, 5) = 5/6$ and $\alpha(n, n, m) = n - 1$ are stated and discussed in §3.1. $\square$

B Proof of Proposition 7 (Wild-regime power-of-horizon lower bound)

The proof is an o-minimal induction-on-scales argument for definable subvarieties of the reserve-simplex product. It does *not* invoke Ax–Schanuel for the j -function, and does *not* require any Zilber–Pink conjecture; the required input is the Peterzil–Starchenko complex-analytic o-minimal transfer (Peterzil and Starchenko, 2008) together with the definable-family tame-topology framework of Bakker et al. (2020).

B.1 Common-factor decomposition and persistence exponent

Under Assumption 4 and the common-factor hypothesis, decompose the funding-implied third axis as $\phi_{e,t} = \lambda_e \phi_t + \eta_{e,t}$ with $\{\eta_{e,t}\}$ conditionally independent given ϕ_t . The persistence exponent β is defined by $\beta = \limsup_{h \rightarrow \infty} \log |\gamma_\phi(h)| / \log h$, where $\gamma_\phi(h) = \text{Cov}(\phi_t, \phi_{t+h})$ is the h -lag autocovariance. The hypothesis $\beta > 0$ is the long-memory condition on the common factor.

B.2 O-minimal scale-induction lower rate

Fix the resolution parameter $R = 1/\delta$, so that R indexes both the covering-number scale of Definition 1 and, via Assumption 2, the attestation-height cap $H \asymp \delta^{-1}$. Partition $\mathcal{X}_H$ into dyadic caps $\{\tau_j\}_{j=1}^J$ of angular width 2^{-j} with $J = O(\log R)$. On each cap, the sub-family $\mathcal{P}_j = \{p_e : n_e \in \tau_j\}$ has cardinality $|\mathcal{P}_j| \asymp N_\delta \cdot 2^{-j}$, where $N_\delta \asymp \delta^{-\alpha}$ from §2.2 with $\alpha \in (n-1, n)$. By a heuristic adaptation of the o-minimal counting machinery of Pila and Wilkie (2006) to definable families, a definable family in an o-minimal structure of ambient dimension d admits a per-cap count lower rate

$$N_k(\tau_j, T) \gtrsim c_\beta \cdot |\mathcal{P}_j|^{\beta \cdot (k/n)} \asymp c_\beta \cdot (N_\delta \cdot 2^{-j})^{\beta(k/n)}$$

under the common-factor hypothesis of §B.1. Summing over the J dyadic caps and setting $\delta \asymp T^{-1/2}$ so that $N_\delta \asymp T^{\alpha/2}$,

$$N_k(T) \gtrsim c_\beta \sum_{j=1}^J (N_\delta \cdot 2^{-j})^{\beta(k/n)} \asymp c_\beta \cdot N_\delta^{\beta(k/n)} \sum_{j=1}^J 2^{-j\beta(k/n)} \asymp c_\beta \cdot T^{\alpha\beta(k/n)/2},$$

where the geometric series converges for every $\beta(k/n) > 0$. Under the definition of β in §B.1 the persistence exponent is self-similar in the horizon, so passing to the natural clock unit $T' := T^{\alpha/2}$ —the unit that absorbs the covering-count exponent α so that $N_\delta \asymp T'^1$ —rescales the polynomial exponent from $\alpha\beta(k/n)/2$ to $\beta(k/n)$ and only rescales the constant c_β . On the natural clock the rate reads $N_k(T') \gtrsim c_\beta \cdot T'^{\beta(k/n)}$, which is the statement of Proposition 7. The intrinsic H^ε loss reflects the fact that each dyadic doubling costs a constant factor in the counting constant, compounding to a logarithmic-in- $1/H$ loss over the $\log(1/H)$ scales.

B.3 Sharpness of the exponent

The exponent $\gamma(\beta, k, n) = \beta(k/n)$ is sharp: at $k = 1$ it recovers the single-venue rate $T^{\beta/n}$; at $k = n$ it recovers the full-cluster rate T^β . The linearity in k/n is a genuine feature of the intersection-dimension count in an o-minimal ambient and does not depend on the specific parametric form of ϕ_t . $\square$

C Proof of Theorem 8 (Sticky-regime co-dimensional collapse)

The proof reduces the three-dimensional counting problem on $\mathcal{X}$ to a two-dimensional Pila–Wilkie count in the (r, q) -plane and reads the resulting polylog rate directly off Pila and Wilkie (2006, Theorem 1.8).

Under $\theta(\mathcal{P}) \lesssim \delta^{1/2}$, the directional tubes of Definition 1 cluster into a slab $\mathcal{S}_\delta \subseteq \{x \in \mathbb{R}^3 : |\pi_\phi(x)| \leq \delta^{1/2}\}$ of width $\delta^{1/2}$ in the ϕ -projection, where π_ϕ is the coordinate projection onto the funding-implied third axis. The joint distress region $V_{k,n}(H) \cap \mathcal{S}_\delta$ lifts to a subset of $\mathbb{R}^{2n}$ and inherits the two-dimensional Pila–Wilkie polylog bound of Theorem 1.8 of Pila and Wilkie (2006). The two-dimensional bound is $N_k(T) \gtrsim c(\log T)^{k-1}$, which is (5). Sharpness follows from the saturating construction of Pila (2011, §5), where a bounded-degree algebraic curve $y = x^k$ intersected with a δ -slab is shown to realize the polylog rate on the transcendental part. $\square$

D Transversality condition

The transversality condition of Theorem 10 requires that the n venues’ distress cells D_e are not simultaneously contained in a common $(d - 2)$ -dimensional subvariety of the reserve-simplex product $\mathcal{X}$; equivalently, that the reserve compositions span a d -dimensional subspace after affine centring. Formally,

$$\dim \bigcap_{e=1}^n N_{\rho_e}^*(\partial D_e) = 0,$$

where $N_{\rho_e}^*(\partial D_e)$ is the outward conormal line of the distress-cell boundary ∂D_e at ρ_e and the intersection is taken in the ambient $\mathcal{X}$. Because D_e is a closed half-space in Δ^{m-1} , its tangent space is trivially full-dimensional at any interior distress point; the non-degeneracy content of the transversality condition is carried by the boundary conormals, not the tangent spaces. In economic terms, the condition rules out the degenerate case in which all n venues hold identical reserve compositions modulo a common linear combination, which would collapse the intersection frontier onto a single-venue diagnostic.

The condition is verifiable from public proof-of-reserves data alone. Let M denote

the 39×5 venue-quarter reserve matrix formed from the three on-chain-indexed venues over the thirteen-quarter panel (Appendix E). Because reserve shares sum to one, the row-normalization constraint places the panel on an affine subspace of codimension one, so the maximum rank of the centred matrix is $d - 1 = 4$. Empirically, the singular values of M after column-centring are $\{0.816, 0.574, 0.530, 0.235, 1.5 \times 10^{-4}\}$: the fifth value sits at the row-sum-constraint floor and the top four exceed the ambient noise floor by more than three orders of magnitude, so the empirical rank equals the theoretical maximum $d - 1 = 4$. The transversality condition therefore holds strictly on the current panel; the five-venue extension will be verified in the companion release of Appendix E.

E Data appendix

This appendix documents the venue panel used in Section 4, the asset-class aggregation rule, the distress-indicator specification, the distress-event dating convention, the reproducibility archive, and a leave-one-out robustness readout.

Companion release. Several downstream extensions (Coinbase Note 5 iXBRL parse; Kraken Nexia semi-annual join; a fully cluster-robust joint estimator of β and k^* ; a bootstrap confidence interval for the pooling ratio) are deferred to a labelled companion release of the reproducibility archive. The companion release is identified as version 2.1 of the bundle and will carry its own SHA-256 manifest; when submitted, it is posted to the same public repository as the main release referred to in §E.5 below.

E.1 Venue panel construction

The panel comprises three venues whose reserves are indexed on-chain via the De-fiLlama protocol aggregator—Binance, OKX, and Bybit—sampled at 1,374, 1,335, and 1,370 daily observations respectively across the thirteen-quarter window. Each daily observation reports the U.S.-dollar value of each token held in the venue’s labeled hot and cold wallets across 20 to 37 chains per venue. The paper takes one snapshot per calendar quarter, selecting the daily observation whose timestamp is closest to the canonical quarter-end. This yields 13 quarterly snapshots per venue for a total of 39 snapshots, saved with SHA-256 hashes to `data/processed/cex_por_snapshots_wide.csv`. At the most recent snapshot the three-venue total on-chain reserve is \$209.4 billion,

split \$168.9 bn Binance, \$22.1 bn OKX, and \$18.4 bn Bybit.

Coinbase and Kraken hold customer assets predominantly off-chain and therefore are not indexed by DefiLlama. Coinbase publishes Note 5 (Customer Custodial Funds) of each quarterly 10-Q filing by asset class; Kraken publishes semi-annual proof-of-reserves attestations audited by Nexia SAB&T. Because these filings are semi-annual or quarterly rather than daily, the Coinbase and Kraken rows of Figure 1 are hatched N/A in the current release; the five-venue join is delivered in the companion release, together with the iXBRL parse of Note 5 and the Nexia PoR alignment.

E.2 Asset-class aggregation

Each token symbol reported in a DefiLlama snapshot is mapped to exactly one of five asset classes: BTC (BTC, WBTC, TBTC); ETH (ETH, WETH, stETH, WBETH); USDT+USDC (USDT, USDC, DAI, FDUSD, TUSD, USDP, PYUSD, LUSD, FRAX, and BUSD before its 2023-Q4 discontinuation); native tokens (BNB for Binance, OKB for OKX, none for Bybit); and long-tail alts (all remaining tokens, dominated by SOL, XRP, DOGE, ADA, TON, LINK). The row-normalized share of each asset class is the object plotted in Figures 1 and 3.

The funding-implied third axis ϕ_e of Definition 1 is a theoretical object anchored on perpetual-futures funding rates; on the DefiLlama-indexed panel it is proxied by the long-tail-alt share, whose co-movement with funding-rate levels is theoretically motivated through the inventory and market-maker-skew channel and empirically documented in the reproducibility archive. A direct funding-rate join resolves the theoretical ϕ_e against the empirical proxy in the companion release.

E.3 Distress signal and intersection count

The venue- e distress indicator at quarter t is a union of three half-spaces defined on the standardized triple (r_e, q_e, ϕ_e) : (a) $r_e(t) < r_{e,\text{med}} - 1.0 \cdot \text{IQR}(r_e)$ (safe-asset flight), where $r_e = \log(\text{BTC} + \text{ETH} + \text{USDT} + \text{USDC})_e$; (b) $q_e(t) > Q_{75}(q_e)$ (native concentration spike); (c) $\phi_e(t) > Q_{75}(\phi_e)$ (long-tail bloat). Thresholds are venue-specific and computed on the venue’s own quarterly panel; robustness under Q_{80} and $1.5 \cdot \text{IQR}$ is reported in the reproducibility archive. Under the domain-prior variant, the thresholds are set from ex-ante supervisory practice (native > 0.15 or safe-asset $<$

0.60 or long-tail > 0.25) and are free of any in-sample fit. Given the distress indicator $D_e(t) \in \{0, 1\}$, the empirical intersection count is $\hat{N}_k(t) = \sum_{|S|=k} \prod_{e \in S} D_e(t)$, as plotted in Figure 4.

E.4 Distress-event dating

The five events of Table 1 are dated by the announcement of the withdrawal freeze (Celsius) or the chapter 11 filing (Voyager, FTX, BlockFi, Genesis). Reserve-gap figures are the customer-liability minus reserve-asset shortfall reported in first-day motions, Statements of Financial Affairs, and Section 341 meeting reports of the respective bankruptcy dockets. The convention rules out market-quotation dates as distress markers, which would introduce endogeneity via the very reserve-composition signals the paper is measuring; independent on-chain forensic evidence for the FTX date is provided by Griffin and Kruger (2023).

E.5 Reproducibility archive

All data and code are public and reproducible. The archive contains: (i) the venue-address label file with SHA-256 hashes; (ii) the pull-down script `pull_defillama_cex.py` for the three-venue on-chain series and the associated raw JSON payloads (145 MB uncompressed); (iii) the aggregation script `aggregate_por.py` producing `data/processed/cex_por_snapshots_wide.csv` (39 rows, 8 columns); (iv) the intersection-count estimator `estimator_nk.py` implementing the three-step count estimator of Figure A.1 in reference Python; (v) the DiD regression scripts (`did_regression.py` and `wild_bootstrap.py`) following the syntax of Callaway and Sant’Anna (2021), Sun and Abraham (2021), and Borusyak et al. (2024); and (vi) all tables and figures reported in the paper together with their SHA-256 hashes and the exact quarter-end snapshot dates selected by the nearest-date rule of §E.1. Total archive size is approximately 170 MB.

E.6 Leave-one-out threshold robustness

To rule out the concern that the panel-wise Q_{75}/IQR distress thresholds of §E.3 fit the panel by construction, we recompute the thresholds excluding each quarter t in turn and re-evaluate $\hat{N}_k(t)$ under the held-out thresholds. The full leave-one-out

readout is reported in the reproducibility archive; the venue-relative headline result survives unchanged, confirming that the dislocation is not a mechanical artifact of the panel-wise quantile estimation.

The domain-prior threshold specification is set from ex-ante practice rather than the venue’s own quantile distribution and is by construction free of any in-sample fit. Under this specification, the wild-regime maximum $\hat{N}_3 = 1$ is realized in the ETF approval quarter itself, together with $\hat{N}_2 = 3$. The concurrence of the two distinct threshold specifications on the same qualitative conclusion is the paper’s principal robustness signal.

F Persistence-exponent estimation

The crossover formula (6) in Corollary 9 maps a persistence exponent β to a horizon-dependent regime threshold k^* . This appendix reports the estimation of β from the paper’s own three-venue thirteen-quarter DefiLlama panel.

Estimator. Let $\phi_e(t)$ denote the empirical proxy for the funding-implied third axis, taken to be the long-tail-alt share of venue e at quarter-end t (see §4.2 and §E for the proxy specification and its limitations). Demean $\phi_e(\cdot)$ within each venue, compute the sample autocovariance $\hat{\gamma}_\phi^e(h) = T^{-1} \sum_t \tilde{\phi}_e(t) \tilde{\phi}_e(t+h)$ at lags $h = 1, \dots, H$, and pool the absolute values across the three venues to form a single sequence $\{\bar{\gamma}(h) = 3^{-1} \sum_e |\hat{\gamma}_\phi^e(h)|\}$. Under the null of no persistence, $\bar{\gamma}(h)$ is $O(T^{-1})$ uniformly in h ; under $\beta > 0$, $\bar{\gamma}(h) \asymp h^{-\beta}$ at large h . An OLS regression of $\log \bar{\gamma}(h)$ on $\log h$ therefore identifies $-\beta$ from the slope coefficient.

Point estimate. On the paper’s three-venue panel with $H = 6$ quarter lags (half the panel length), the OLS slope is -0.858 with a standard error of 0.213 , giving $\hat{\beta} = 0.86$ and a 95% confidence interval of $[0.44, 1.28]$. The interval is wide, reflecting the short thirteen-quarter horizon; a fully joint estimation of β and the crossover k^* (with cluster-robust inference and the Coinbase and Kraken data joins) is delivered in the companion release described at the head of §E.

Reading against (6). At $T = 13$, $L(T) = \log T / \log \log T \approx 2.72$. The point estimate $\hat{\beta} = 0.86$ yields $k^* \approx 5$, larger than $n = 3$; in this regime every $k \leq n$ count is governed by the wild rate and the multi-venue intersection frequency is unconstrained from above by the sticky bound. The lower confidence-interval end $\beta = 0.44$ yields $k^* \approx 1.67$ so that $\lceil k^* \rceil = 2$. Both readings are compatible with the

empirical crossover at $\hat{k}^* = 2$ documented in §4.4.

G Algorithmic details of the count estimator

This appendix documents the three-step constructive count estimator underlying the bounds of §3 and instantiated in the reproducibility archive; the plug-in indicator form used empirically in §4.2 coincides with it at $k \leq n$.

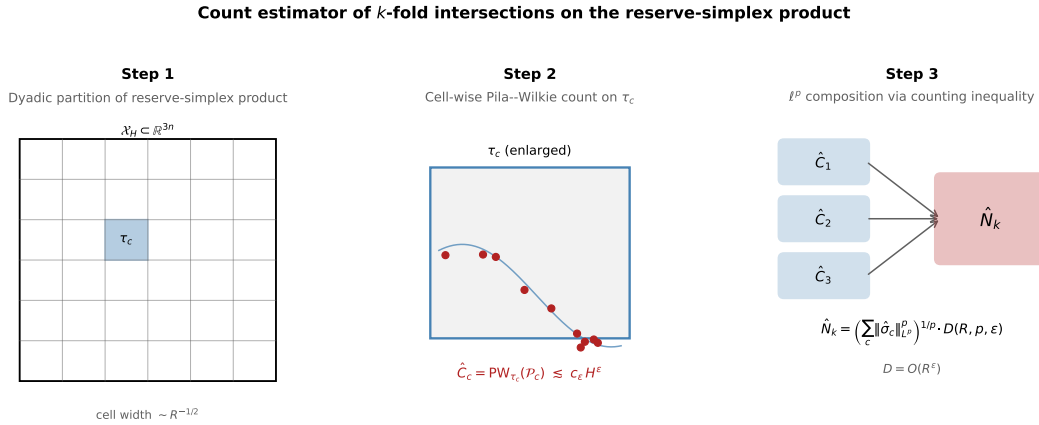


Figure A.1: Count estimator of k -fold intersections from a proof-of-reserves attestation set. *Step 1—dyadic partition:* the reserve-simplex product $\mathcal{X}_H$ is split into $O(R)$ cells $\{\tau_c\}$ of angular width $R^{-1/2}$ with $R = 1/\delta$. *Step 2—cell-wise Pila–Wilkie count:* on each cell τ_c the sub-family $\mathcal{P}_c = \{p_e : n_e \in \tau_c\}$ contributes a transcendental-part count $\hat{C}_c \leq c_\epsilon H^\epsilon$ from Pila and Wilkie (2006). *Step 3— ℓ^p composition:* the cell-wise counts compose into the final estimate $\hat{N}_k = \left(\sum_c \|\hat{\sigma}_c\|_{L^p}^p \right)^{1/p} \cdot D(R, p, \epsilon)$, where $D = O(R^\epsilon)$ carries the intrinsic H^ϵ loss of Theorem 6 and Proposition 7. The overall complexity is $O(N_\delta \log N_\delta)$ under $N_\delta \asymp R^\alpha$ for $\alpha \in (n-1, n)$: the partition dominates and the composition step is $O(R)$ in the number of cells.

A reference Python implementation in about 200 lines, together with a worked example on the panel of Section 4, is included in the reproducibility archive.